3.6.2 Thermal physics (A-level only)

3.6.2.1 Thermal energy transfer (A-level only)

Content	Opportunities for skills development
Internal energy is the sum of the randomly distributed kinetic energies and potential energies of the particles in a body. The internal energy of a system is increased when energy is transferred to it by heating or when work is done on it (and vice versa), eg a qualitative treatment of the first law of thermodynamics. Appreciation that during a change of state the potential energies of the particle ensemble are changing but not the kinetic energies. Calculations involving transfer of energy. For a change of temperature: $Q = mc\,\Delta\,\theta$ where c is specific heat capacity. Calculations including continuous flow. For a change of state $Q = ml$ where l is the specific latent heat.	**MS 1.5 / PS 2.3 / AT a, b, d, f** Investigate the factors that affect the change in temperature of a substance using an electrical method or the method of mixtures. Students should be able to identify random and systematic errors in the experiment and suggest ways to remove them. **PS 1.1, 4.1 / AT k** Investigate, with a data logger and temperature sensor, the change in temperature with time of a substance undergoing a phase change when energy is supplied at a constant rate.

3.6.2.2 Ideal gases (A-level only)

Content	Opportunities for skills development
Gas laws as experimental relationships between p, V, T and the mass of the gas. Concept of absolute zero of temperature. Ideal gas equation: $pV = nRT$ for n moles and $pV = NkT$ for N molecules. $Work\ done = p\Delta V$ Avogadro constant N_A, molar gas constant R, Boltzmann constant k Molar mass and molecular mass.	
Required practical 8: Investigation of Boyle's law (constant temperature) and Charles's law (constant pressure) for a gas.	**MS 3.3, 3.4, 3.14 / AT a**

3.6.2.3 Molecular kinetic theory model (A-level only)

Content	Opportunities for skills development
Brownian motion as evidence for existence of atoms. Explanation of relationships between p, V and T in terms of a simple molecular model. Students should understand that the gas laws are empirical in nature whereas the kinetic theory model arises from theory. Assumptions leading to $pV = \frac{1}{3}Nm(c_{\mathrm{rms}})^2$ including derivation of the equation and calculations. A simple algebraic approach involving conservation of momentum is required. Appreciation that for an ideal gas internal energy is kinetic energy of the atoms. Use of *average molecular kinetic energy* = $\frac{1}{2}m(c_{\mathrm{rms}})^2 = \frac{3}{2}kT = \frac{3RT}{2N_A}$ Appreciation of how knowledge and understanding of the behaviour of a gas has changed over time.	